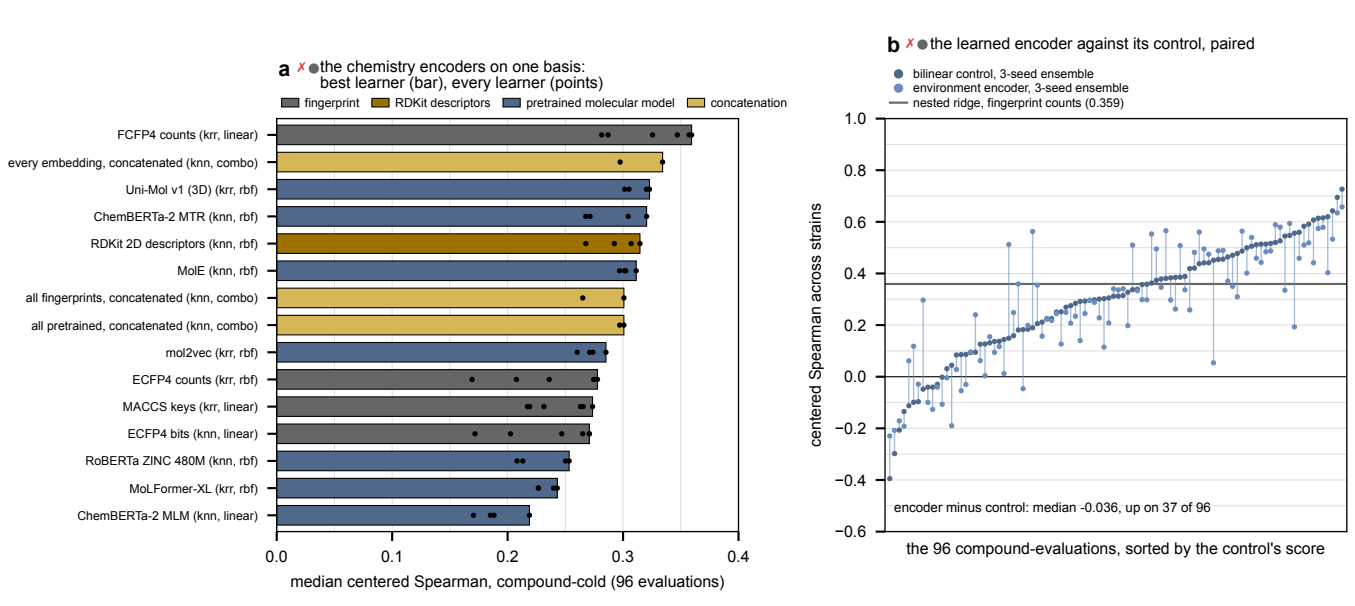

a ✗ ● the chemistry encoders on one basis:
best learner (bar), every learner (points)
fingerprint
RDKit descriptors
pretrained molecular model
concatenation
FCFP4 counts (krr, linear)
every embedding, concatenated (knn, combo)
Uni-Mol v1 (3D) (krr, rbf)
ChemBERTa-2 MTR (knn, rbf)
RDKit 2D descriptors (knn, rbf)
MolE (knn, rbf)
all fingerprints, concatenated (knn, combo)
all pretrained, concatenated (knn, combo)
mol2vec (krr, rbf)
ECFP4 counts (krr, rbf)
MACCS keys (krr, linear)
ECFP4 bits (knn, linear)
RoBERTa ZINC 480M (knn, rbf)
MoLFormer-XL (krr, rbf)
ChemBERTa-2 MLM (knn, linear)
0.0
0.1
0.2
0.3
0.4
median centered Spearman, compound-cold (96 evaluations)
b ✗ ● the learned encoder against its control, paired
bilinear control, 3-seed ensemble
environment encoder, 3-seed ensemble
nested ridge, fingerprint counts (0.359)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
centered Spearman across strains
encoder minus control: median -0.036, up on 37 of 96
the 96 compound-evaluations, sorted by the control's score